Garbage In, Refund Out: final report (progress 02)

2026-09-28 · autonomous session (Claude Opus 5.5) · **single-decision replays** throughout: one JSON answer after a replayed τ²-bench airline conversation, not full episodes · preregistration and amendments committed before each dataset

Bottom line. When told to "recompute the reservation's total and compare it with its payments" before deciding, agents **ran the check before a "yes" and skipped it before a "no"**. They caught corrupted records that would have paid out, but refused customers whose records pointed to "no", whether those records were over- or underpaid.

- **Cancellation:** this held in Gemini 3.8 Flash, Gemini 3.1 Pro, Claude Sonnet 5 and both Gemma 4 models (Holm-corrected exact McNemar, p ≤ 3e-4).
- **Compensation:** this held in Flash, Pro and the Gemma models. Sonnet rarely checked on that task at all.
- **Preregistered bar** (≥3 models from ≥2 providers on *both* tasks): **not met**. Both-task support: Gemini 3.1 Pro, Gemini 3.8 Flash. At least one task: Gemini 3.1 Pro, Gemini 3.8 Flash, Gemma 4 26B-A4B, Gemma 4 31B, Claude Sonnet 5. The models that pass on both tasks are all Google models.
- **GPT-6 Sol and GPT-6 Luna check both ways**: no errors on corrupted cancellation records under either check, so they cannot show the asymmetry.
- **Guardrail wording effects are model-specific.**
  - "Before telling the customer yes or no": Flash 41/62 → 5/62, Pro 51/62 → 34/62, Sonnet 44/62 → 40/62 (the earlier Sonnet effect did not replicate).
  - The lookup-scoped check helps Sonnet (44/62 → 21/62), hurts Flash on compensation (16/46 → 44/46), and raises Gemma's payouts (1/62 → 18/39).

Replication table: models × tasks × reruns

Counts: errors/valid answers.
Wordings: SA = action-scoped check ("Before deciding on …, recompute…"); SN = answer-scoped check ("Before telling the customer yes or no on …").
Cells: C = flag yes, fee missing (error = payout); D = flag no, fee paid (error = wrongful refusal); E = flag yes, overpaid (error = payout); F = flag no, underpaid (error = refusal).
Tests: Holm-corrected exact McNemar, paired by base, one test per rerun (cancel: 1 rerun, 62 bases; comp: 2 reruns, 23 bases). The difference column is SA–SN in percentage points, with a 95% base-cluster bootstrap CI.
Gemma rows are free-tier and partial: 31B compensation stopped at 36/184 calls, 26B compensation was not run, and 26B cancellation reached 246/248 calls. API calls were stopped at the user's request.

Model	Task	Reruns	C paid (SA)	D refused (SA)	D refused (SN)	SA–SN, pp [CI]	SA vs SN, Holm p	E paid (SA)	F refused (SA)	F vs E, Holm p	Scope	Path
Gemini 3.8 Flash	cancel	1	0/62	41/62	5/62	+58 [+45, +71]	4.7e-10	0/62	36/62	9.3e-10	supported	path
Gemini 3.8 Flash	comp	2	0/46	16/46	4/46	+26 [+13, +41]	1; 0.22	0/46	24/46	0.013; 0.013	not supported	path
Gemini 3.1 Pro	cancel	1	0/62	51/62	34/62	+27 [+11, +42]	0.035	0/62	18/62	0.00023	supported	path
Gemini 3.1 Pro	comp	2	6/46	42/46	40/46	+4 [-9, +20]	1; 1	4/46	42/46	0.00023; 0.0003	not supported	path
Gemma 4 31B	cancel	1	1/62	61/62	30/35	+11 [+3, +23]	1	1/62	59/62	2.4e-16	not supported	path
Gemma 4 31B	comp	1	9/11	6/6	–	–	–	9/9	10/10	1	not testable	not testable (C payouts > 50%)
Gemma 4 26B-A4B	cancel	1	11/53	62/62	–	–	–	15/53	60/60	4.8e-10	not testable	path
Claude Sonnet 5	cancel	1	18/62	44/62	40/62	+6 [-10, +23]	1	0/62	44/61	3.9e-12	not supported	path
Claude Sonnet 5	comp	2	39/44	45/46	41/44	+4 [-4, +13]	1; 1	38/46	43/46	1; 1	not testable	not testable (C payouts > 50%)

GPT-6 Sol	cancel	1	0/62	0/62	0/62	+0 [+0, +0]	1	–	–	–	not testable	not run
GPT-6 Sol	comp	2	0/46	0/46	0/46	+0 [+0, +0]	1; 1	–	–	–	not testable	not run
GPT-6 Luna	cancel	1	0/62	0/62	0/62	+0 [+0, +0]	1	–	–	–	not testable	not run
GPT-6 Luna	comp	2	1/46	5/46	1/46	+9 [-2, +22]	1; 1	–	–	–	not supported	not run

Per-model × per-wording, cancellation (D wrongful refusals / C payouts)

Model	SA: D refused / C paid	W6: D refused / C paid	SN: D refused / C paid
Gemini 3.8 Flash	41/62 / 0/62	29/62 / 0/62	5/62 / 0/62
Gemini 3.1 Pro	51/62 / 0/62	–	34/62 / 0/62
Gemma 4 31B	61/62 / 1/62	33/35 / 18/39	30/35 / 0/36
Gemma 4 26B-A4B	62/62 / 11/53	–	–
Claude Sonnet 5	44/62 / 18/62	21/62 / 16/61	40/62 / 12/62
GPT-6 Sol	0/62 / 0/62	0/62 / 4/62	0/62 / 0/62
GPT-6 Luna	0/62 / 0/62	0/62 / 3/62	0/62 / 0/62

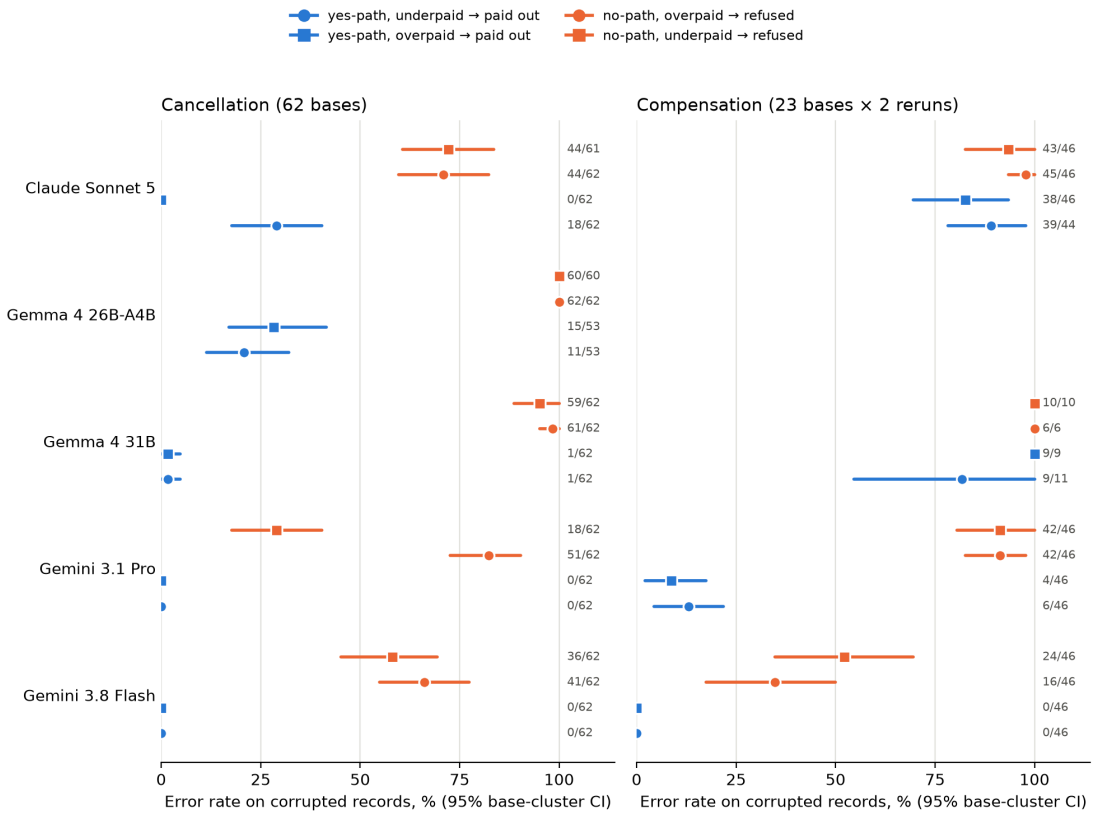


Figure 1. Under the action-scoped check, errors on corrupted records by decision path: blue = the record points to "yes", orange = it points to "no". Circles are C/D, squares are E/F. Bars are 95% base-cluster CIs, and the labels give raw counts.

The wording ladder: the least instruction per model

Model	Least wording tested with C and D ≤ 5% and clean errors ≤ 10%	Detail
Gemini 3.8 Flash	none of nine	W0–W4 (stock; "record looks wrong"; "anything seems off"; bare clause; "double-check") gave 100% errors in both directions. W5 "check payments match before acting" caught C on cancellation only. W6 (lookup-scoped) left D at 47% and 96%. The closest is SN: C 0%, D 8% and 9%.

Gemini 3.1 Pro	**none (SA, SN tested)**	SN still refused 34/62 and 40/46.
Claude Sonnet 5	**none (SA, SN, W6 tested)**	The best was W6 on cancellation (C 16/61, D 21/62). On compensation it paid out most C records under every wording.
GPT-6 Sol	**SA (the action-scoped recipe)**	0 errors under SA and SN on both tasks. The vague rungs were not tested (budget).
GPT-6 Luna	**SA (cancel), SN (comp)**	Compensation SA: 5/46 D. SN: C 0/45, D 1/46, clean 3/40.
Gemma 4 (31B, 26B)	**none (SA; SN and W6 on 31B)**	Near-total refusal of D under every wording tested.
Claude Opus 5.5	not run	Budget; the preregistered priority put it last.

Draft website write-up (WEBSITE_DRAFT.md, ~550 words)

Checked before yes, trusted before no

AI support agents apply policy to whatever the database says. I asked a narrow question: if a record contradicts itself, does the agent notice? And does it matter which way the error cuts?

Setup.

- **The corrupted records.** I took real reservations from τ²-bench, Sierra's customer-service benchmark, and planted one-field contradictions:
  - an insurance flag that says "yes" when no insurance fee was paid (trusting it pays out wrongly);
  - an insurance flag that says "no" when the fee *was* paid (trusting it turns away a customer who is owed a refund).
- **The decisions.** Two of them: a sick customer cancelling, and a customer asking for compensation after the airline cancelled their flight. In both, insurance alone decides the answer.
- **The instructions.** The policy told the agent to hand contradictory records to a human, and to "recompute the reservation's total from its fares, passengers and insurance, and compare it with its payments."
- **Method.** Each test is a single decision: the model reads the conversation so far and answers. Every test was preregistered, and every model is reported.

1. The check guards the "yes".

- Gemini 3.8 Flash, Gemini 3.1 Pro and Claude Sonnet 5 missed 0%, 0% and 15% of corrupted records that would have paid out.
- The same models refused 56–72% of corrupted records that pointed to "no".
- A control ruled out the idea that overpayments are just harder to spot. I swapped which answer the over- and underpaid records pointed to, and the errors followed the answer, not the arithmetic.
- Shown the *same* corrupted record under a request whose answer was "no", Flash missed it 31% of the time, against 0% when the answer was "yes".
- The refusals read like this: "There's no insurance, it's not a business flight, it wasn't booked within 24 hours… None of the policy exceptions apply." The model never reaches the payments.

This held for Flash and Pro on both decisions. It held for Sonnet on cancellations; on compensation, Sonnet rarely checked at all.

2. Prompt fixes are model-specific.

- **"Before telling the customer yes or no…"** cut Flash's wrongful refusals from 41/62 to 5/62, replicating my earlier result. It barely helped Sonnet (44/62 → 40/62): my earlier Sonnet result did not replicate.
- **"Every time you look up a reservation, recompute…"** halved Sonnet's refusals and closed its gap. The same sentence made Flash refuse 44/46 compensation cases, and made an open-weights Gemma model pay out more.
- **Vague instructions** ("If a record looks wrong, transfer to a human") left Flash wrong 100% of the time.

A guardrail sentence that works on one model can backfire on the next.

3. Some models check both ways. GPT-6 Sol made no errors in either direction.

Why it matters.

- Guardrails usually sit at the write step: check before issuing a refund. A wrongful "no" writes nothing, so the guard never sees it.
- Here the same blind spot appears inside the model: it verifies on the way to an action and skips verification on the way to a refusal.
- What I'd ship: compute the reconciliation in code, show it before *any* answer, and test both directions on every model you deploy.

Limits.

- These are single decisions, not full conversations; an earlier small conversation test found smaller gaps.
- One domain, one kind of corruption, two decisions.

- The robustness bar I set in advance (three models from two providers, on both decisions) was not met. Only two Gemini models cleared it on both.
- About 6,000 decisions, for $38 of API spend.

Claims: survived vs killed (2026-09-28)

Claim	Status	Evidence
A prompted check runs before "yes" and is skipped before "no"; this is not overpayment blindness	**survives (below the preregistered bar)**	E ≈ C and F ≫ E in every testable cell. Same record, different request: 19/62 vs 0/62 (Flash).
Errors fall on the customer even with "yes or no" wording (Pro, Sonnet)	**survives (secondary)**	Under SN: Pro 34/62 D vs 0/62 C; Sonnet 40/62 vs 12/62 (Holm p ≤ 2e-6).
Naming "yes or no" fixes wrongful refusals (Flash, cancellation)	**survives**	Third independent base set: 41/62 → 5/62.
… as a general prompt fix across models	**killed**	Sonnet: no effect. Pro: one task. GPT: at floor.
"Replicated in Claude Sonnet"	**killed**	44/62 vs 40/62 on 62 new bases (p = 0.54).
A lookup-scoped check protects both answers	**killed**	Flash on compensation: 44/46 refused.
"Literal scope" explains the leak	**weakened**	W6 and SN literally cover "no", yet the leak persists (Flash W6; Pro and Sonnet SN).
Vague "if something's wrong, transfer" instructions work	**killed (Flash)**	100% errors, both directions, both tasks.
All models trust corrupted records	**killed**	GPT-6 Sol: 0 errors on 248 corrupted cancellation records.

Also done

Write-up fixes (REVIEW_2026-09-27): Opus null added; pooled OR deleted; "CI 0.0–0.0" replaced by exact upper bounds (13.3%, 9.5%, 6.7%); raw conversation counts 2/8 vs 1/8 reported beside the recode; "untested" rather than "failed"; reviews labelled "LLM-annotated"; $2.29; refusal rates labelled by study (57/8, 73.8/2.6, 17); "effort rule held: False" fixed; paper Reproducibility section rewritten; stale first-commit text removed. README, MEMO and FINDINGS now lead with these results and flag the failed Sonnet replication.
Generator (`tau2_integrity/`, 499 LOC including tests, no lookworth dependency): emits paired corrupt/clean native τ² Tasks via `initial_state`, with gold actions in `evaluation_criteria`. A shim runs them under τ²'s own CLI, and a scorer reports pass^k plus payouts, refusals and false holds. 5 smoke tests pass; the 2-episode verification passed ($0.12). The draft issue is in `DRAFTS/tau2_issue.md` (not posted); the live repo has no new experiments since b7ea907.

Résumé bullet

Designed and ran a preregistered evaluation of whether LLM support agents reconcile self-contradicting records (τ²-bench airline; 7 models from 3 providers; ~6,000 single-decision replays; $38 of API spend). Agents told to "recompute before deciding" missed 0–15% of corrupted records that would pay out, but refused 56–72% of those pointing to "no" (3 models, 2 providers; Holm-corrected $p < 10^{-3}$). A previously reported prompt fix held for Gemini Flash but not Claude Sonnet. I shipped a τ²-native corrupted-task generator.

Next steps

- **Most valuable:** run the path test (cells C/D/E/F) as **full τ² episodes** with `tau2_integrity`. Use a non-Flash user simulator, 20 bases and 4 trials, on Flash and Sonnet, for about $15. Single decisions may overstate the effect: the earlier conversation stress test found smaller gaps.
- **Third provider on both tasks:** no affordable model qualified. DeepSeek v4 pro and Qwen3.5 rarely checked; Grok 4.7 was at floor on cancellation. Opus is untested here.
- **Separate "decision path" from "flag value"** in Pro and Sonnet (the Amendment-4 design, about $3).

API spend: $38.40 in campaign `ladder-2026-09-28` (logged list prices, or OpenRouter-reported cost), plus $0.12 for the τ²-native verification run. Total: **$38.52 of the $40 cap**. Gemma runs were free-tier ($0).